



5

RESOURCES IN LEARNING MATHEMATICS

- 5.1 *Introduction*
- 5.2 *Learning Outcomes*
- 5.3 *Importance of Resources in Primary Mathematics Learning*
- 5.4 *Contextual and Inclusive Use of Resources*
- 5.5 *Critical Analysis of Mathematics Textbooks and Supplementary Materials*
- 5.6 *Let Us Sum Up*
- 5.7 *End Exercises*
- 5.8 *References*

5.1 INTRODUCTION

Mathematics learning at the primary stage forms the foundation for higher conceptual understanding and problem-solving skills. As you've seen in the earlier units, we've focused on **child-centered pedagogy** and the significance of **foundational numeracy**—because learning mathematics is like a spiral. Now, it's time to take the next step.

This unit invites you to **go beyond the fundamentals**—to explore how we can make mathematics **meaningful, joyful, and rooted in the child's world**. Together, we'll explore the importance of using **manipulatives**, integrating **local and digital resources**, and designing **cross-curricular projects** that bring mathematics **alive** for the students, with the objective of nurturing **curious and, confident problem-solvers**.

Understanding of
Pedagogy and
Resources

So, are you ready to transform your math classroom into a space of discovery and connection? Let's begin!

**5.2 LEARNING OUTCOMES**

After completion of the chapter, the learner:

- understands the pedagogical role of diverse resources in primary mathematics;
- designs, develops and documents the use of contextual, inclusive, and age-appropriate mathematics learning resources;
- appreciates the relevance of socially and culturally responsive mathematics learning and apply it;
- plans the integration of mathematical concepts across the curriculum by designing meaningful projects; and
- critically analyses mathematics textbooks and supplementary materials.

5.3 IMPORTANCE OF RESOURCES IN PRIMARY MATHEMATICS LEARNING

Teaching mathematics by confining it to the bounds of textbook-based instruction often leads to limited understanding and disinterest in the subject. Concrete, semi-concrete, and abstract modes of representation—using **various resources**—support conceptual clarity and help in developing interest, particularly for young learners. Resources help make mathematics relatable, tangible, and visible.

Teaching mathematics solely through textbook-based instruction often results in rote learning, superficial understanding, and a growing disinterest in the subject, especially among young learners. To nurture genuine mathematical thinking, it is essential to move beyond procedural teaching and offer learners experiences that engage multiple senses and representations.

The Concrete–Semi-concrete–Abstract (CSA) approach provides a developmental pathway for deep understanding. In this model, students first interact with concrete manipulatives, then transition to semi-concrete (pictorial) representations, and finally develop the ability to engage with abstract symbols and formal notation (Bruner, 1966). This scaffolded process is particularly

significant in primary education, where children's cognitive abilities are still evolving.

Research consistently emphasizes that using diverse resources—manipulatives, story cards, interactive games, real-life objects, and technological tools—makes mathematical ideas tangible and relatable (Clements & Sarama, 2014). Manipulatives help in bridging the gap between abstract concepts and children's lived experiences, promoting both conceptual clarity and sustained engagement.

This approach is firmly rooted in cognitive constructivist theories, which emphasize active learning and knowledge construction through meaningful experiences. A substantial body of research over the decades has consistently highlighted the importance of using concrete resources in mathematics education, particularly at the primary level. In this section, we will examine the role of resources in enhancing mathematical understanding among young learners, grounding the discussion in the foundational ideas of cognitive constructivism.

5.3.1 Importance of Resources Through A Theoretical Lens

Jean Piaget (Piaget, 1952) and Lev Vygotsky (Vygotsky, 1978) both highlighted the value of practical, hands-on experiences in mathematics education, particularly in early and middle childhood. According to their theories, a key component of successful learning is the use of resources, especially manipulatives, which are tangible items used to teach mathematical concepts.

Both Jean Piaget (1952) and Lev Vygotsky (1978) emphasized the importance of hands-on, concrete experiences in learning mathematics, especially during early and middle childhood. Their theories support the use of resources, particularly manipulatives—physical objects used to teach mathematical concepts—as a crucial part of effective learning.

According to Piaget, primary school children largely fall under the Concrete Operational Stage (7-11 years). Children begin to think logically about concrete events and can classify, do series and understand conservation. He believed that learning must match the child's developmental stage, hence emphasizing the importance of concrete materials (manipulatives) for this stage. Manipulatives help children internalise abstract mathematical concepts like number sense, arithmetic operations, volume, and area by bridging the gap between concrete experience and abstract thinking.

Children at this age learn best when they are given the opportunity to actively participate in the process of learning. Engagement with the material, they can touch and manipulate, helps them think critically and develop fundamental

**Notes**

Understanding of Pedagogy and Resources



Notes

numeracy skills like observing patterns, classification, identifying and creating series, to name a few, as per their developmental readiness.

However, according to Vygotsky (1978), manipulatives are like “**meditating tools**” that support learning. When utilized with the guidance from teachers and peers, manipulatives help children achieve the tasks that are beyond their cognitive stage but fall under their zone of proximal development (ZPD).

Vygotsky’s Zone of Proximal Development (ZPD)

Imagine a child learning to ride a bicycle

In the beginning, a child may not be able to balance the bicycle and hence can’t ride the bicycle on its own. This is the task that has to be learned.

However, the child is comfortable riding a tricycle on its own. This is the task that the child has mastered.

Then think of a situation where a child is learning to ride a bicycle with supporting wheels. That’s where the magic happens!

This is what Lev Vygotsky called the Zone of Proximal Development (ZPD).

The ZPD is the gap between:

What a learner can do independently

and

What they can do with guidance or collaboration

Vygotsky believed that learning happens best in this zone, when a more knowledgeable person—like a teacher, peer, or parent—offers **scaffolding** (guidance and support) like a training wheel.

Illustrative example:

Let’s say you’re teaching multiplication of fractions in class 5.

Task 1: Multiplying whole numbers? Achievable! Students can do it on their own. (This is below the ZPD—already mastered.)

Task 2: Multiplying fractions by fractions? Not achievable ! This could be challenging right now, as this concept may not be a part of their previous knowledge and mastery of the concept. (This is above the ZPD—it has to be mastered.)

Task 3: But if you work together with your students and use meditating tools to deliver the concept, like using a pizza model, the students slowly begin to get the conceptual clarity. They ask questions, make mistakes, and correct themselves—and you guide them through it all. (This is the ZPD in action!)

Over time, what was once “with help” becomes “I can do it myself”—the student’s learning gets evolved.

Vygotsky (1978) emphasized that language and dialogue around the use of manipulatives are essential for concept formation. He emphasizes the social context of the learner as well. Hence implying the importance of folklore, indigenous stories and real-life experiences of the learner in supporting conceptual understanding of mathematics.

Both theorists support the use of manipulatives in mathematics education, but Piaget (1952) emphasizes developmental readiness, while Vygotsky (1978) stresses social context and scaffolding. Together, their theories advocate for concrete, interactive, and supported learning experiences in mathematics.

Similarly, Maria Montessori (Montessori, 1967) placed hands-on learning at the very heart of her educational philosophy. She believed that children learn best through active, self-directed exploration using concrete materials specially designed to support independent learning.

Montessori’s Views on Hands-On Learning:

1. “The hand is the instrument of the mind”: She emphasized that learning occurs through movement and manipulation. Sensorial and motor experiences help internalise abstract concepts.
2. Concrete to Abstract Progression: Children begin with concrete, tactile materials (e.g., number rods, bead chains, golden beads) before moving to symbolic or abstract understanding. This aligns with how children naturally construct knowledge.
3. Prepared Environment: Children can select and use the carefully arranged materials in Montessori classrooms on their own. These resources enable practice and mastery and are self-correcting.
4. Growth of Self-Sufficiency and Concentration: Active learning promotes independence, focus, and self-motivation. Instead of lecturing, teachers serve as mentors, letting kids learn by doing.
5. Children benefit from Montessori (1967) math resources such as the stamp game, spindle boxes and the golden bead material for place value.

**Notes**

Understanding of Pedagogy and Resources



Notes

Montessori math materials (like the golden bead material for place value, spindle boxes, or the stamp game) help children:

1. Visualise and physically manipulate numbers
2. Understand concepts such as place value, operations, and fractions through concrete experience
3. Build a deep conceptual foundation before moving to symbolic representation

Montessori championed hands-on learning as essential for

- Engaging both the mind and body
- Supporting natural development
- Laying a strong foundation for abstract thinking

Her approach aligns with developmental principles and continues to influence progressive education practices worldwide.

Several other theorists and educators support the use of resources and manipulatives in primary school mathematics. Here are a few notable ones:

Jerome Bruner (Bruner, 1966) – Theory of Representational Stages

Three Modes of Representation:

- Enactive (action-based)—Learning through physical manipulation (e.g., blocks, beads).
- Iconic (image-based)—Using pictures or visual models.
- Symbolic – Using abstract symbols like numbers or letters.

Bruner (1966) advocated that children should first experience concepts physically (enactive) before moving to pictorial and symbolic forms—making manipulatives essential in early math education.

Zoltan Dienes (Dienes, 1960)—Theory of Variability and Manipulatives

Advocated using multiple representations and manipulatives to teach mathematical concepts. Emphasized “structured games” and “perceptual variability”—learning concepts through different forms and contexts to deepen understanding.

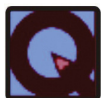
Developed several teaching tools (e.g., Dienes blocks) to teach place value and operations concretely.

Seymour Papert (Papert, 1980)—Constructionism

A protégé of Piaget, Papert (1980) extended constructivism into constructionism—learning best occurs when learners actively construct meaningful products.

Advocated the use of concrete tools and technology (e.g., the Logo programming language) to help children explore mathematical ideas.

We can conclude, from Piaget, Vygotsky, Montessori, Bruner, Dienes, and Papert collectively that resources and manipulatives are not just supplementary tools but essential elements of effective mathematics learning. They support learners' transition from concrete to abstract thinking, support their developmental stages, encourage exploration, and foster collaborative learning. By actively engaging with hands-on materials, learners build a deeper, more meaningful understanding of mathematical concepts, laying a strong foundation for lifelong mathematical thinking, which is more meaningful and enjoyable.



CHECK YOUR PROGRESS 5.1

1. Why should mathematics instruction at the primary level move beyond textbook routines to include diverse resources and materials? What risks do we face if we neglect this dimension of teaching?
2. In what specific ways can manipulative-based learning transform a child's engagement with mathematical concepts? Consider at least three advantages, but also reflect critically—are there limitations or conditions under which manipulatives might not serve their intended purpose?
3. Piaget (1952) emphasizes the role of concrete operational thinking in children's cognitive development. How does this theory validate the use of hands-on manipulatives in mathematics teaching? Can we also interrogate the boundaries of Piaget's assumptions in today's varied classroom contexts?
4. Vygotsky (1978) places learning within a social and cultural framework. How does his theory of the Zone of Proximal Development encourage the use of shared, resource-based mathematical explorations? Further, how might the teacher's role shift when learning is mediated by tools, peers, and collaborative dialogue?



Notes

Understanding of Pedagogy and Resources



5. Name one limitation of using manipulatives in math learning.
6. Choose the correct option:
 - (i) Why is it important to move beyond textbook-based routines in primary mathematics teaching?
 - (a) Textbooks cover all possible learning contexts
 - (b) Diverse materials make learning mechanical
 - (c) Textbook routines often limit conceptual understanding and engagement
 - (d) Textbooks are the only validated resources
 - (ii) Which of the following is NOT an advantage of manipulative-based learning?
 - (a) It helps children visualise and experiment with concepts.
 - (b) It encourages passive note-taking.
 - (c) It supports multiple learning styles.
 - (d) It promotes active engagement
 - (iii) According to Piaget (1952), during the concrete operational stage, children learn best when:
 - (a) They memorize formulas by rote.
 - (b) They use hands-on materials to manipulate ideas.
 - (c) They only listen to lectures.
 - (d) They are assessed through written tests only.
7. Fill in the Blanks
 - (i) Using only textbook-based instruction in primary mathematics can lead to _____ and a superficial understanding of mathematical ideas.
 - (ii) Manipulatives like blocks, counters, and fraction strips help children move from _____ to abstract mathematical thinking.
 - (iii) Piaget's theory may not fully address the needs of today's diverse classrooms because it assumes a _____ path of cognitive development for all children.

Answers:

6. (i) (c) (ii) (b) (iii) (b)
7. (i) disengagement (ii) Concrete (iii) Universal or linear

5.3.2 Types of Resources for Teaching Mathematics

Effective primary school mathematics teaching and learning processes necessitate the careful integration of a range of resources that engage students, put abstract ideas into context, and encourage active, meaningful learning. These materials, which can be digital, visual, auditory, or physical, should be in line with the experiences and developmental stages of the learners. Curiosity, critical thinking, and a favorable attitude towards mathematics are all encouraged by a rich fusion of traditional and modern resources.

In this section we will walk you through various types of resources along with practical examples. These examples are not exhaustive; you are encouraged to explore and discover additional resources based on your learners' needs and the content-specific requirements.

1. Concrete and Manipulative Resources

Children can move, touch, and experiment with these tangible items. Mathematical concepts such as number sense, operations with numbers, geometry, and measurement can be visualized with the aid of manipulatives. Discussed below are a few available resources:

The Magic Box, or Jadui Pitara: This resource pack, which was created by NCERT as part of the NIPUN Bharat Mission, contains vibrant, tactile materials such as puzzles, number cards, bead strings, and measuring tapes. For instance, kids can practice counting, skip counting, or basic addition and subtraction with a Jadui Pitara bead string.



Fig 5.1 Jadui Pitara (Source: <https://ejaadupitara.ncert.gov.in/>)



Notes

Understanding of Pedagogy and Resources



Jodo Gyan Kit: Developed by Jodo Gyan, this kit includes innovative manipulatives like Ganit Mala (a number necklace), Rangometry (geometric shapes), and place value cards. These materials help demystify abstract concepts and link math to daily life. For example, the Ganit Mala can help children internalize the number line and visualize addition as “jumps.”

Arvind Gupta Toys: Using simple, recycled materials, Arvind Gupta’s toys transform everyday items into engaging mathematical tools. A matchstick geometry set or spinning tops to demonstrate symmetry are both affordable and effective.

2. Print Resources

Books, story cards, charts, and workbooks are traditional but powerful resources when used thoughtfully. They support not only procedural learning but also help develop mathematical language, reasoning, and comprehension (Van de Walle et al., 2013). Research shows that mathematical storybooks and contextual problems can enhance children’s conceptual understanding by linking abstract concepts to real-life situations (Anderson et al., 2005). Well-chosen print materials encourage learners to visualize problems, engage in reflective thinking, and foster connections across mathematical ideas (National Council of Teachers of Mathematics [NCTM], 2014).

Discussed below are some quality print materials:

NCERT Textbooks (2025 edition): These new textbooks are designed with a focus on experiential learning. Each chapter begins with a context (e.g., a fair, a train journey) and includes embedded activities, stories, and games. For instance, Class 3’s chapter “Give and Take” introduces the concept of money with the help of grocery purchases.

The Man Who Counted by Malba Tahan: A story-based approach to math that stimulates imagination while reinforcing concepts. Stories like “The Horse Problem” bring in division and problem-solving in a fun way. This narrative-based method of teaching mathematics develops fundamental ideas while igniting students’ interest and creativity. For example, the humorous story “The Caterpillar Who Went to School” presents the idea of measurement. Children interact with the concept of units and lengths through storytelling as the caterpillar learns how to measure itself and others, giving math a purpose and a sense of enjoyment. Children also learn about area and measurement in the story “The Clever Tailor,” where the tailor ingeniously stitches a complete garment out of leftover fabric.



Notes

Such stories not only bring math alive through real-world contexts but also help children connect numeracy with logic, creativity, and cultural wisdom.

Mathematwist: Number Tales from Around the World by T.V. Padma uses engaging folk tales and stories from different cultures to introduce and reinforce mathematical concepts in a fun and imaginative way.

Posters and Wall Charts: Visual displays of multiplication tables, geometric shapes, or math vocabulary serve as constant reinforcement tools.

3. Games and Recreational Mathematics

In casual contexts, games encourage participation, teamwork, and the use of mathematical reasoning.

Sudoku: Promotes the ability to recognize patterns, reason logically, and arrange numbers.

Snakes & Ladders and Ludo: Excellent for practicing counting, moving forward and backward, and comprehending chance in younger children.

Hopscotch (Stapu): A classic game that can be modified to teach addition, subtraction, and number sequencing. Children jump on numbered squares, gaining motor coordination and numeric fluency simultaneously.

Math Bingo or Number Tambola: Fun ways to reinforce number recognition, multiples, or even decimal values.

Playing Cards: A deck of cards can not only help children in developing early probability thinking but also develop skills like pattern recognition and critical thinking.

4. Rhymes, Songs, and Folk Stories

Mathematical concepts can be internalized deeply when set to rhythm or narrated through stories, especially for early primary grades.

Number Rhymes: Songs like “One, two, buckle my shoe...” introduce counting and sequence in pre-primary classes.

Folk Tales and Stories: Tales involving sharing among animals or distributing sweets in villages can introduce concepts like division or estimation.

Classroom-created songs: Teachers and students together can create rhymes for concepts like shapes (“I am a triangle, I have three sides...”) or measurements.

Understanding of Pedagogy and Resources



5. Digital and Technological Resources

In the digital era, interactive tools and apps provide new dimensions to math learning. There are many AR and VR applications that are free to use. Some significant ones are shared below:

DIKSHA App and Portal: Provides teacher-developed content, lesson plans, activities, and videos aligned with the NCERT curriculum. For instance, interactive videos on fractions or measurement from DIKSHA can be used to supplement classroom teaching.

Desmos: A number line on Desmos is highly interactive, where children can explore various operations like addition and subtraction.

GeoGebra: Helps visualize geometry concepts through drag-and-drop, transformations, and constructing figures.

Interactive Whiteboards and Smart Class Content: Certain digital learning platforms provide engaging modules that animate mathematical processes. As teachers, we need to first critically analyze the relevance of these platforms to use them judiciously with our students.

6. Environmental and Contextual Resources

Using elements from the child's environment is both cost-effective and meaningful due to the familiarity of the child.

Stones, Leaves, Matchsticks: For counting, patterning, sorting, or forming shapes.

Local Markets: A real-world setting to learn about money, weight, estimation, and data collection.

Classroom Objects: Desks, books, and erasers—used for teaching measurement, grouping, or comparison.

Math Walks: Engage children in observing and recording numbers and shapes in their surroundings—for example, room numbers, classroom labels, or the number of steps in a staircase. During this activity, teachers can highlight the difference between **cardinal numbers** (used for counting) and **ordinal numbers** (used for positioning) as children gather data about classroom and room allocations. A math walk in the school garden or open area allows children to explore and note the shapes and numbers found in nature and infrastructure—such as the arrangement of trees, the pattern of parallel bushes, the shapes of manhole covers, and various signs.



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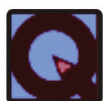
7. Teacher-Created and Student-Made Resources

Involving children in creating math tools fosters ownership and creativity. This addresses the problem of alienation many children experience in mathematical learning.

Flashcards, handmade dice, and angle testers can be designed and used by children to reinforce learning. For example, each child can create their own angle tester and move around to identify angles all around them.

Student Projects: Making models of 3D shapes, math posters, or board games integrates math with art and craft.

Math Journals: Children can create a math journal to record mathematical data about themselves, their body, bodies families, their addresses, and their neighborhoods. This helps in personalizing mathematics and is often seen as a strategy to develop fondness for the subject.



CHECK YOUR PROGRESS 5.2

1. Think of an example for each resource type from your classroom experience, or suggest any new idea that you think you could use in the future:
 - a. Concrete Manipulative —
 - b. Thematic Kit —
 - c. Digital Resource —
2. Choose the Correct Option:
 - (i) Why is it important to integrate both traditional and modern resources in primary mathematics classrooms?
 - (a) To cover more content quickly
 - (b) To align with national assessment frameworks
 - (c) To connect abstract concepts with children's real-life experiences and foster curiosity
 - (d) To replace textbooks entirely
 - (ii) Which of the following best reflects the purpose of using manipulatives like *ginladi* or bead strings?
 - (a) To help children memorize tables
 - (b) To visualize abstract concepts and support active learning through hands-on experience

Understanding of Pedagogy and Resources



Notes

- (c) To keep students occupied with physical activity
- (d) To prepare children for competitive exams
- (iii) How do teacher-created and student-made resources help address math anxiety or alienation?
 - (a) By reducing the syllabus load
 - (b) By providing repetitive drill practice
 - (c) By giving students ownership and connecting math to personal experiences
 - (d) By focusing only on easy math problems

3. Fill in the blank:

- (i) The Jadui Pitara is a resource developed by _____ as part of the NIPUN Bharat Mission.
- (ii) Involving children in creating resources like flashcards, handmade dice, or angle testers fosters _____ and reduces the sense of alienation from mathematics.
- (iii) **Math Walks** help children differentiate between _____ numbers used for counting and _____ numbers used for positioning.

Answers:

- 2. (i) (c) (ii) (b) (iii) (c)
- 3. (i) NCERT (ii) Ownership and creativity (iii) Cardinal, ordinal

5.4 CONTEXTUAL AND INCLUSIVE USE OF RESOURCES

In the pedagogy of primary school mathematics, the contextual and inclusive use of resources is central to building meaningful, equitable, and engaging learning experiences. Contextual use refers to embedding mathematics teaching within the lived realities of children—drawing on their cultural practices, language, environment, and everyday activities to make abstract concepts tangible and relatable. For example, using local items like grains, bangles, or traditional patterns in rangoli for teaching counting, measurement, and geometry helps bridge the gap between home and school learning. Similarly, using local currency, familiar objects (such as mangoes and marbles), or community-based scenarios (like market scenes or festivals) can help children visualize and internalize concepts like addition, measurement, and pattern recognition.

Understanding of
Pedagogy and
Resources

Inclusive use ensures that all children, regardless of ability, language, gender, or socio-economic background, have access to and can benefit from mathematical learning. The Indian educationist Krishna Kumar (1992) has argued for the use of culturally responsive pedagogy to make education more relevant and humane, especially in diverse classrooms.

Policy frameworks in India have increasingly emphasized inclusive and equitable approaches to mathematics education. The National Curriculum Framework (NCF) 2023 and the National Curriculum Framework for Foundational Stage (NCF-FS) 2022 foreground experiential, play-based, and competency-driven learning that connects mathematical understanding to real-life contexts. These frameworks align with the vision of the National Education Policy (NEP) 2020, which calls for equitable, inclusive, and flexible education that caters to diverse learners, including children with disabilities and those from disadvantaged groups. Earlier, the National Curriculum Framework (NCF) 2005 had laid the foundation for such an approach by emphasizing contextual and meaningful mathematics through the use of local, low-cost materials that promote exploration, participation, and understanding.

By integrating local, culturally appropriate, and low-cost materials and adapting resources for children with special needs (such as tactile materials, large print charts, or audio tools), teachers can transform the mathematics classroom into an inclusive, dynamic learning space. Such practices not only promote conceptual clarity and engagement but also affirm each child's identity, enabling them to see themselves as valued learners of mathematics.

Inclusivity in resource use ensures that materials cater to the diverse needs of all learners, including those with different learning styles, abilities, languages, and socio-economic backgrounds. This involves the use of multi-sensory aids such as tactile materials, audio-visual content, manipulatives, and pictorial representations to support diverse learners. For children with disabilities, customized tools such as abacuses with tactile markings or audio instructions can be used. Teachers must also be sensitive to gender, caste, and cultural biases in resource selection, ensuring that no group is marginalized. By integrating locally available, low-cost, and culturally appropriate teaching-learning materials, teachers not only enhance students' mathematical understanding but also foster a sense of identity, belonging, and confidence in the learning process. This approach aligns with the constructivist pedagogy and principles of Universal Design for Learning (UDL), making mathematics education more accessible, relevant, and enjoyable for every child.

Universal Design for Learning (UDL): Making Math Work for Every Learner

As educators, we've all seen how students engage with mathematics differently—some thrive on patterns and procedures, while others need to *see* or *touch* the



Notes

Understanding of Pedagogy and Resources



Notes

concept to really “get it.” This is exactly where **Universal Design for Learning (UDL)** comes in—not as an extra layer of planning, but as a mindset shift that helps us build more flexible and inclusive math classrooms.

So, what *is* UDL?

Think of UDL as designing your math resources and lessons from the start to reach **all** learners—not just by adding accommodations after the fact. It’s about offering **multiple ways to present content**, **multiple ways for students to show what they know**, and **multiple ways to keep them engaged**.

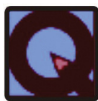
How does this look in math?

- **Representation:** Ever noticed how visual aids like number lines or fraction bars help some students suddenly “click” with a concept? UDL encourages us to bring in visuals, manipulatives, and even tech tools (like virtual interactive geometry software) to offer different entry points for understanding.
- **Action & Expression:** Not every student performs well in a timed ‘paper and pen’ test. What if some explained their reasoning verbally, created visual proof, or used a math app to demonstrate understanding? The UDL approach would promote this—let’s give learners those options.
- **Engagement:** We know how motivation matters. Contextualizing problems (say, using sports data, architecture, or budgeting problems) can engage learners who might otherwise lose focus due to alienation of the context. UDL encourages us to build in more choices and challenges that keep the learner engaged.

Why does it matter?

UDL is about designing resources that work across diverse needs *before* students get stuck. It’s a powerful way to promote **equity**, **accessibility**, and **deeper conceptual understanding** without lowering expectations or compromising academic rigor.

In short, UDL is a smarter, more inclusive way to do what we already do best: *help every student grow mathematically.*



CHECK YOUR PROGRESS 5.3

1. Reflect on your current teaching practices. Can you think of an example where a mathematical concept could be taught using a real-life scenario that is relevant to your students’ local culture or everyday life? How might you modify resources or strategies to include learners with different abilities, languages, or learning styles?



Notes

2. Think about the concepts you are currently teaching—such as data handling, measurement, money, or geometry. How might a trip to the local market provide opportunities for hands-on exploration of these topics? What activities or follow-up tasks could make the learning deeper and more meaningful for your students?
3. Choose the correct option:
 - (i) What is the primary goal of using contextual resources in primary mathematics classrooms?
 - (a) To cover the syllabus faster
 - (b) To make mathematical concepts relatable and meaningful
 - (c) To replace textbooks entirely
 - (d) To focus only on rote memorization
 - (ii) Which of the following is an example of integrating mathematics with environmental studies?
 - (a) Drawing symmetrical patterns using Rangoli
 - (b) Memorizing multiplication tables
 - (c) Tracking plant growth and recording measurements
 - (d) Solving equations without real-life connections
 - (iii) Universal Design for Learning (UDL) encourages teachers to:
 - (a) Provide a single method of teaching to maintain uniformity
 - (b) Design resources from the start to meet diverse learners' needs
 - (c) Use only digital tools in teaching
 - (d) Avoid using real-life examples to prevent bias

Answers:

3. (i) (b) (ii) (c) (iii) (b)

5.4.1 Integration of Mathematical Concepts Across The Curriculum

Integration of mathematical concepts across the curriculum is a powerful pedagogical strategy in primary school education, promoting holistic learning and deeper conceptual understanding. Mathematics, far from being an isolated subject, naturally connects with other disciplines such as environmental studies, language, art, and physical education. Through interdisciplinary learning, children begin to see mathematics as a tool for making sense of the world around them, rather than as a set of abstract rules and procedures.

Understanding of
Pedagogy and
Resources

Notes

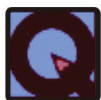
For example, in environmental studies, children can explore measurement by recording rainfall or tracking plant growth, thereby applying mathematical concepts like units, data collection, and graphing. Language lessons offer opportunities to reinforce mathematical vocabulary (e.g., “more than,” “equal to”) and encourage reasoning and communication of ideas. Art activities like creating symmetrical patterns, tessellations, or Rangoli designs allow learners to apply geometry and spatial understanding in creative ways. In physical education, children can engage with numbers, counting, estimation, and time while participating in games and exercises. These experiences help in reinforcing mathematical thinking in meaningful, context-rich ways.

The theoretical underpinning for this integrated approach lies in the constructivist paradigm advocated by John Dewey (Dewey, 1938), who emphasized learning through experience and the interdependence of knowledge domains. Howard Gardner (Gardner, 1983)’s Theory of Multiple Intelligences also supports this approach by recognizing that children learn in diverse ways—some through movement, some through visuals, and some through language—highlighting the need for integrated, multi-sensory learning experiences.

From a policy perspective, the National Curriculum Framework (NCF) 2023 and the National Curriculum Framework for Foundational Stage (NCF-FS) 2022 highlight the importance of integrated, multidisciplinary, and experiential learning that bridges subject boundaries and strengthens foundational numeracy. These frameworks build on the vision of the National Education Policy (NEP) 2020, which promotes 21st-century competencies and interconnected learning experiences. Earlier, the National Curriculum Framework (NCF) 2005 had stressed breaking rigid subject divisions and promoting cross-curricular linkages so that mathematics learning remains close to the child’s reality and integrates experiences from other domains. Together, these policy documents reaffirm the importance of connecting mathematics with real-life contexts and diverse knowledge systems to nurture holistic, meaningful, and inclusive learning experiences.

Integrating mathematics across the curriculum not only enhances learning outcomes but also makes mathematics more enjoyable, relevant, and meaningful. It encourages collaborative learning, critical thinking, and application of knowledge in real-life situations—skills essential for nurturing confident, problem-solving learners. By thoughtfully using resources from various subject areas, teachers can foster a rich, inclusive mathematical environment that supports diverse learners and promotes holistic development.

Cross-curricular integration promotes holistic understanding. Mathematics connects naturally with environmental studies, language, art, and physical education. Supported by John Dewey (Dewey, 1938) and Howard Gardner (Gardner, 1983)’s theories, and NCF 2005 and NEP 2020.

**CHECK YOUR PROGRESS 5.4**

Think about the subjects you teach or collaborate on—where do you already see natural overlapping with the mathematical concepts? Can you design an activity or a project (e.g., a science experiment or a map-reading task) where students apply mathematical concepts like number operations, measurement, data analysis, or patterns in an authentic context?

5.4.2 Socially and Culturally Integrated Mathematics Learning

Socially and culturally integrated mathematics learning emphasizes the use of teaching-learning resources that reflect the lived realities, cultural knowledge, and social contexts of learners. Drawing on the work of scholars such as Eric Gutstein (Gutstein, 2006), Bob Peterson (Peterson, 2005), Brian Greer (Greer, 1996), and Swapna Mukhopadhyay (Mukhopadhyay, 2009), this approach challenges the traditional, decontextualized teaching of mathematics and instead promotes an inclusive, empowering, and socially conscious pedagogy.

Gutstein (2006) and Peterson (2005), through their work on Teaching Mathematics for Social Justice, advocate for resources that enable students to critically examine real-world issues—such as economic inequality, environmental sustainability, and access to public services—using mathematical tools. For example, classroom resources can include charts comparing water consumption across neighborhoods, surveys on school facilities, or infographics on gender disparities in education. Such data-driven tasks help children apply mathematical concepts like statistics, averages, and ratios to analyze social conditions and develop critical consciousness.

Brian Greer (Greer, 1996) and Swapna Mukhopadhyay (Mukhopadhyay, 2009) highlight the importance of ethnomathematics—the study of mathematical ideas present in cultural practices. They argue that every community has embedded mathematical knowledge in its daily life, from local architecture and handicrafts to traditional games and measurement systems. Incorporating these elements as classroom resources not only grounds learning in familiar contexts but also affirms the cultural identity of learners. For instance, using patterns from local textiles to teach symmetry or exploring indigenous farming calendars to understand measurement and time introduces mathematics in a way that is relevant and respectful of cultural diversity.

This approach also supports student agency and voice, as emphasized by Gutstein (Gutstein, 2006). By involving learners in posing problems, investigating their communities, and collecting real-life data, teachers can transform students



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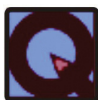
Understanding of Pedagogy and Resources



from passive recipients of knowledge into active participants in the learning process. Resources created or adapted in collaboration with students—such as neighborhood maps, family budgets, or school surveys—enhance engagement, ownership, and critical thinking.

Incorporating socially and culturally responsive resources aligns closely with the principles articulated in the National Curriculum Framework (NCF) 2023 and the NCF-FS 2022, both of which advocate experiential, competency-based, and culturally grounded learning across stages of schooling. These frameworks, together with the National Education Policy (NEP) 2020, emphasize equity, inclusion, and the integration of indigenous knowledge systems. They build upon the vision of the National Curriculum Framework (NCF) 2005, which encouraged the use of child-centered and locally relevant materials. Collectively, these policy documents reflect the constructivist belief that children learn best when new knowledge connects to their existing experiences and contexts.

By integrating mathematics into the social and cultural life of students, teachers can not only deepen conceptual understanding but also promote capable, thoughtful human beings who view mathematics as a significant and powerful aspect of their own world.



CHECK YOUR PROGRESS 5.5

1. What is ethnomathematics? Reflect on the diverse cultural practices within your classroom community. Are there local art forms, traditional games, or community practices that involve mathematical thinking—like patterns, measurements, or symmetry? How might you use these to help students connect their cultural identity to mathematical learning?
2. Think about the social issues that matter to your students—such as water availability, regular power supply/power cuts, access to education, or local demographics. How could you incorporate this data into a math lesson using graphs, percentages, or comparisons? What impact might this have on student engagement and their understanding of both math and the world around them?
3. Choose the correct option:
 - (i) Which scholar is associated with teaching mathematics for social justice?

(a) Eric Gutstein	(b) Howard Gardner
(c) John Dewey	(d) Krishna Kumar
 - (ii) Ethnomathematics focuses on:
 - (a) Teaching mathematics only through textbooks
 - (b) Exploring mathematical ideas embedded in cultural practices
 - (c) Preparing students for competitive exams
 - (d) Ignoring students' social contexts

Answers:

3. (i) (a) (ii) (b)

**5.4.3 Socially and Culturally Responsive Mathematics Teaching:
Examples And Activities**

In order to successfully incorporate culturally and socially appropriate resources into mathematics teaching, classroom teachers can create learning experiences that bridge mathematical ideas with students' cultural and social worlds. The following are some sample activities corresponding to primary school themes:

1. Spatial Concepts (Geometry)

Activity: Have children draw a basic map of their neighborhood, highlighting sites like the local market, park, temple/mosque/church, and their own homes.

Math Concept: Position, direction, distance, and shapes.

Resource Used: Student experiences, community layout.

Extension: Incorporate words such as “left,” “right,” “above,” “below,” and simple units of measurement.

2. Traditional Games for Counting and Strategy (Number Operations)

Activity: Utilize local games such as Pachisi, Stapu (Hopscotch), or Gilli-Danda to practice skip counting, addition, or strategy-based problem-solving.

Math Concept: Number sense, operations, probability.

Resource Used: Cultural games, movement-based learning.

Extension: Encourage students to tabulate scores or create new rules for the game based on mathematics.

3. Patterns in Art and Crafts (Patterns and Symmetry)

Activity: Introduce examples of rangoli, tribal painting, or fabric prints and discuss repeating patterns and lines of symmetry.

Math Concept: Patterns, symmetry, and geometry.

Used Resource: Art forms that are culturally important to children.

Extension: Children can make their own symmetrical patterns using natural or locally available materials.

**Notes**

Understanding of
Pedagogy and
Resources



Fig 5.2 Symmetry and Pattern in Rangoli (Source: AI generated image)

4. Data from Daily Life (Data Handling)

Activity: Take a class survey of subjects such as “modes of transport to school” or “favorite local food” and note down responses.

Math Concept: Collection, representation (tally marks, pictographs), and interpretation.

Resource Used: Class experiences, chart paper, colored markers.

Extension: Use the data to talk about frequency, create bar graphs, or identify the mode.

5. Festivals and Time (Measurement and Calendars)

Activity: Explain how festivals are associated with the lunar/solar calendar and create a simple classroom calendar that signifies local festivals.

Math Concept: Time measurement, sequencing, calendar math.

Resource Used: Cultural calendars, story.

Extension: Students can make individual event timelines based on birth dates, festivals, and family celebrations.

6. Income and Budget (Money and Transactions)

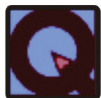
Activity: Utilize scenarios such as shopping at a local market or festival preparation to introduce budgeting.

Math Concept: Money, addition and subtraction, estimation.

Resource Used: Locally priced goods (real or role-play), charts.

Extension: Involve students in a mock “market day” where they play the roles of buyers and sellers using play money.

These activities illustrate that resources need not always be purchased—many can be constructed from the life, culture, and community of the student. As teachers, one should feel supported to listen and observe, using the rich cultural capital that children bring to the classroom. This not only enhances mathematical learning but also constructs confidence, respect, and inclusivity in multi-cultural classrooms.

**CHECK YOUR PROGRESS 5.6**

1. Think about the traditional art forms in your community—such as rangoli, mandala designs, Warli art, or kolam. How might you use these in a math lesson to explore ideas like rotational symmetry, tessellation, or fractions? How would this help students appreciate both mathematical concepts and their cultural roots?
2. Consider festivals your students celebrate. How can you use them to teach days of the week, months, leap years, or sequencing events on a timeline? Could students create festival calendars, calculate intervals between festivals, or explore lunar vs. solar calendars? How would this deepen their understanding of time and cultural diversity?

5.5 CRITICAL ANALYSIS OF MATHEMATICS TEXTBOOKS AND SUPPLEMENTARY MATERIALS

Textbooks and supplementary materials need to be critically evaluated to ascertain that they are effective pedagogical tools. They need to be evaluated based on the curriculum of the class, normally suggested by the apex bodies. An appropriately designed resource can lead to deeper learning, while poorly conceived content may create hindrance in learning and lead to conceptual gaps. NCERT mathematics textbooks for primary classes provide examples that can serve as indicators to guide. In the following section, NCERT textbooks are used as exemplars to illustrate critical evaluation of textbooks.

- Curriculum Alignment

Example: Class III NCERT (Old Edition) – “Fun with Numbers”

The chapter aligns well with the learning outcomes on number sense and place value outlined in the NCF and Learning Outcomes framework. It is important to ensure that each activity clearly connects to specific mathematical goals. For instance, when storytelling elements are used, teachers can enhance learning by identifying the underlying mathematical goals, complemented by focused observation and questioning.



Notes

Understanding of Pedagogy and Resources



- Conceptual Clarity and Progression

Example: Class IV NCERT (Old Edition)—“Building with Bricks”

The chapter introduces patterns, spatial understanding, and multiplication through visuals and local contexts. Visuals in themselves build a concept here; teachers need to carefully analyze the concept such that concept delivery is not compromised. To maximize learning, teachers can facilitate the transition from visual observation to explicit mathematical reasoning, such as identifying concepts of area or symmetry, ensuring conceptual clarity and smooth progression.

- Pedagogical Strategies

Example: Class II – “Counting in Groups”

This chapter effectively introduces grouping, repeated addition, and foundational multiplication through games and everyday settings. It encourages a constructivist approach, allowing students to uncover patterns and make independent observations. In a classroom setting, incorporating additional prompts for peer discussion or forecasting can further enrich inquiry-based learning and collaborative engagement among students.

- Inclusivity and Contextual Relevance

Example: Class V NCERT (Old Edition) – “Smart Charts”

The chapter “Smart Charts” presents data handling through actual contexts such as surveys among friends or market shopping experiences, promoting meaningful learner engagement. In the classroom, teachers can further enhance inclusivity by adapting examples to reflect diverse regional and socio-economic backgrounds, ensuring that all students connect with the local context in an equitable and relevant manner.

- Visual and Linguistic Accessibility

Example: Class I – “Shapes and Space”

The chapter constructively uses familiar objects such as balls, books, and boxes to introduce concepts of spatial awareness through engaging visuals. The age-appropriate illustrations and simple language effectively support comprehension. In the classroom, teachers can ensure clarity by addressing visuals that may allow multiple interpretations and by thoughtfully examining images to maintain inclusivity and avoid any unintended bias related to caste, class, race, or gender.



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- Manipulatives and Concrete Materials Integration

Throughout Classes I–III:

Textbooks tend to recommend utilising objects such as matchsticks, bottle tops, or beads for counting, grouping, or measuring.

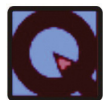
Value: Favors the use of low-cost, easily accessible materials, in line with the views of Piaget (Piaget, 1952) and Bruner (Bruner, 1966) on learning through concrete means.

Note: Trainers may need additional support and time for bringing these recommendations effectively into classroom practice.

NCERT primary school mathematics textbooks are rich in day-to-day contexts, pictures, and child-sensitive language and are good basic sources. However, thoughtful periodic review and reflection are essential so that their implementation becomes effective and inclusive. Teachers need to check regularly:

- The clarity and logical flow of ideas,
- The accessibility and point of relevance of examples,
- The support given to diverse learners.

This reflective practice maximizes textbooks and support materials as effective aids for meaningful mathematics learning in the classroom.

**CHECK YOUR PROGRESS 5.7**

1. When selecting a mathematics textbook for your classroom, what three criteria do you consider most important, and how do they impact student learning and engagement?
2. Reflect on your teaching experience—do you prioritize alignment with curriculum, contextual relevance, inclusivity, or problem-solving focus? How do these criteria affect the way your students interact with the textbook and develop mathematical thinking?
3. Critically analyse the new edition of NCERT Mathematics textbooks I to V (2025). How do these textbooks support or challenge your current teaching practices?
4. Choose the correct option:
 - (i) Why is critical evaluation of textbooks necessary?

Understanding of Pedagogy and Resources



Notes

- (a) To ensure alignment with curriculum goals and promote deeper learning.
 - (b) To reduce the number of chapters
 - (c) To shorten the syllabus for easier teaching
 - (d) To avoid the use of manipulatives
- (ii) What pedagogical suggestion is given for the Class II chapter “Counting in Groups”?
- (a) Reduce the number of activities.
 - (b) Remove games from the chapter.
 - (c) Use only abstract numbers.
 - (d) Add more peer discussion prompts to enhance inquiry
5. Fill in the blank:
- (i) The NCERT primary mathematics textbooks are rich in day-to-day _____, pictures, and child-sensitive language.
 - (ii) Visual representations in textbooks should be checked carefully for possible biases related to _____, _____, _____, or _____.
 - (iii) Teachers need to ensure that every activity in the textbook is connected to a _____ goal.

Answers:

4. (i) (a) (ii) (d)
5. (i) Contents (ii) Caste, class, race and gender (iii) Learning



5.6 LET US SUM UP

Resources are central to meaningful mathematics learning. They must be diverse, inclusive, culturally responsive, and integrated across subjects. The effective use of resources transforms abstract mathematics into an engaging, inclusive, and joyful learning experience.

Developmental theorists support manipulatives and concrete resources in early mathematics because they:

Understanding of
Pedagogy and
Resources

Notes

- Build conceptual understanding
- Support developmentally appropriate learning
- Encourage active exploration and problem-solving
- Help transition from concrete to abstract thinking

When their concepts are combined, a solid theoretical basis for the use of rich, interactive materials in elementary school math classes is created.

- Key theorists emphasize the value of hands-on experiences:
- Piaget (1952): Emphasized the Concrete Operational Stage; advocated for manipulatives that align with developmental readiness.
- Vygotsky (1978): Stressed the role of social interaction and Zone of Proximal Development; advocated guided use of manipulatives.
- Montessori (1967): Promoted independent, hands-on learning with specially designed tactile materials.
- Bruner (1966): Introduced Enactive, Iconic, and Symbolic stages of learning; supported physical experience before abstraction.
- Dienes (1960) and Papert (1980): Advocated variability and meaningful construction through manipulatives and technology.

The selection and use of diverse resources are central to effective mathematics pedagogy at the primary level. A resource-rich environment promotes exploration, experimentation, and confidence in learners. The key is not the number of resources but how creatively and contextually they are used to deepen mathematical understanding. As emphasized in the NEP 2020 and the NCF-FS 2022, foundational numeracy should be built through engaging, inclusive, and joyful learning experiences—made possible through thoughtful resource integration.

- **Concrete and Manipulative Resources:** Jadui Pitara, *ginladi*, Arvind Gupta toys.
- **Print Resources:** Storybooks, charts, and workbooks (e.g., “The Man Who Counted”).
- **Games and Recreational Mathematics:** Hopscotch, Sudoku, and Tambola.

Understanding of Pedagogy and Resources



- **Songs and Rhymes:** Number songs and culturally relevant stories.
- **Digital Resources:** DIKSHA App, Desmos, GeoGebra.
- **Environmental Resources:** Stones, local markets, math walks.
- **Teacher and Student-Created Resources:** Flashcards, journals, and handmade games.

Theorists like Gutstein (2006), Peterson (2005), Greer (1996), and Mukhopadhyay (2009) promote teaching math through social justice, critical thinking, and community data. The pedagogy incorporates learners' cultural and social contexts. Ethnomathematics helps affirm identity and relevance in the learning of mathematics. It recognises that even cooking a dish or baking a cake in the kitchen is equally relevant mathematical skill as the one in the textbook. Such an approach elaborates the domain of mathematical resources and reduces the 'fear' and 'phobia' related to the subject.

Textbooks serve as a major learning resource, in certain circumstances the only learning source. Hence, textbooks should be used judiciously by the teachers after critical evaluation. Evaluation Criteria can be broadly summed up as following:

- Curriculum alignment
- Concept clarity
- Pedagogical strategy
- Inclusivity
- Visual and linguistic accessibility
- Differentiated learning
- Integration of manipulatives

NCERT textbooks offer several good practices but require contextual adaptation. Teachers must use critical lenses to adapt and optimize use.

- They help concretise abstract concepts and support developmental learning.
- Promote active learning, foster engagement, and build confidence.
 - (a) Jadui Pitara
 - (b) *ginladi*
 - (c) DIKSHA App



5.7 END EXERCISES

1. Discuss the role of culturally responsive pedagogy in mathematics.
2. Design a math activity based on a local game.
3. Critically evaluate a Class III NCERT math chapter using given criteria.
4. Create a cross-curricular project integrating math and art.

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UNIT - 2

Understanding of Pedagogy and Resources



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